

Eduardo Vasconcelos

Security Engineer | PhD student

Linux kernel, Android security, AI red teaming, systems programming

Contact

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- **GitHub:** <https://github.com/vasconcedu>
- **GitLab:** <https://gitlab.com/vasconcedu>

Key Highlights

- Almost 10 years of experience in Security Engineering with emphasis on Android;
- Active contributor to the Linux kernel with 3 patches contributed to upstream;
- MSc in Computer Science awarded for national-level excellence in Software Engineering research;
- Computer Engineer awarded for excellence in the undergraduate thesis;
- Experience in teaching Python programming in vocational/technical education;
- Proficient in three languages: Portuguese (*native*), English (*IELTS: C1*), and French (*TCF: B2–C1*).

Skills

- **Advanced:** security engineering, threat modeling, security testing, security code review, R&D
- **Intermediate:** reverse engineering, DevSecOps, cryptography, Linux sysadmin, Linux kernel development, Machine Learning
- **Familiar:** incident response, embedded systems

Work History

- **Master Application Security Architect @ SiDi, Brazil (remote)**

Aug 2024–Present

- Reverse engineered Android malware, decompiling DEX to Java and analyzing x86 shared objects to develop malicious behavior heuristics and Machine Learning models for malware detection;
- Designed and implemented automated Android malware analysis tools in Python to unveil and monitor malicious application behavior;
- Participated in internal AI red teaming task forces, contributing to AI security testing knowledge building and reporting security vulnerabilities in AI agents;
- Delivered AI security training to resident tech students, including AI security engineering and red teaming;
- Performed threat modeling and security testing of Android and web applications, including Samsung Knox;
- Proponent and key contributor in revitalizing the Security Champions program.

- **Open Source Contributor @ Linux kernel**

Apr 2026–Present

- Contributed a total of 3 patches to the upstream Linux kernel;
- Currently focusing on the Linux Security Module (LSM) framework and AppArmor.

- **Technical Instructor @ SENAI, Brazil (on-site)**

May 2025–Jan 2026

- Taught Python programming classes.

- **Senior Mobile Security Engineer @ iFood, Brazil (remote)**

Feb 2022–Aug 2024

- Designed and implemented security and fraud prevention mechanisms in C++, integrated into mission-critical Android applications via Java Native Interface (JNI), serving over 40 million monthly users;
- Reverse engineered malicious Android apps, decompiling DEX to Java and analyzing x86 shared objects, to reveal methods and behaviors used by fraudsters in food delivery services;
- Integrated Runtime Application Self-Protection (RASP) solutions into GitLab CI/CD pipelines for mobile apps.

- **Senior Software Security Analyst @ SiDi, Brazil**

Apr 2019–Feb 2022

- Delivered ongoing offensive security testing services to Samsung, emphasizing Android applications and Android OS system services;
- Conducted research on Android API and OS vulnerabilities on behalf of Samsung, including issues related to Android accessibility services abuses and the permissions system circumvention.

- **Information Security Analyst @ Hacker Rangers, Brazil**

Jun 2017–Apr 2019

- Delivered consulting services in web application and network security assessments and phishing campaigns;
- Developed and operated the Hacker Rangers SaaS security awareness training platform;
- Administered IBM Security Guardium virtual appliances built on Red Hat Linux, handling deployment, configuration, and operation for private organizations and government agencies.

- **Information Technology Intern @ Embraer, Brazil**

Sep 2016–Jun 2017

- Contributed to the development of an IT Change Management plan;
- Developed Python tools to analyze corporate web traffic.

- **Engineering Intern @ SigMEDIA, Ireland**

May 2014–Jul 2014

- Developed a Machine Learning ensemble model to detect laughter in human speech;
- Processed digital voice signals in Python.

Education

- **Doctoral Degree in Computer Science @ USP, Brazil**

Apr 2026–Present

- Research topic: fuzz testing embedded systems operating in brownout voltage regimes for security bugs;
- Currently serving as a volunteer Debian Linux system administrator at the Software Engineering Lab.

- **Master's Degree in Computer Science @ USP, Brazil**

Feb 2022–Aug 2025

- Graduated with an A average;
- Research topic: leveraging Mutation Testing techniques to inject vulnerabilities into Android apps, providing a novel approach to evaluating mobile security testing procedures;
- Received the Distinguished Paper Award at the Brazilian Conference on Software 2024 (CB-SOFT'24) for my work, "Mutation Testing to Support the Security Testing of Android Applications";
- Served as a teaching assistant for Software Engineering.

- **Postgraduate Diploma in Software Engineering @ Unicamp, Brazil**

Feb 2019–Dec 2019

- Graduated with a GPA of 9.4 out of 10.0;
- One-year Postgraduate Diploma program in Software Engineering.

- **Engineer's Degree in Computer Engineering @ USP, Brazil**

Feb 2012–Jul 2017

- Graduated 13th out of 59 with a GPA of 7.5 out of 10.0;
- Awarded the Sao Paulo Research Foundation (FAPESP) undergraduate research scholarship, contributing to a research project focused on flood forecasting in urban areas;
- Received the Best Computer Engineering Undergraduate Thesis Award in 2016 for my thesis on emotion recognition in human speech using Machine Learning;
- Served as a teaching assistant for Python programming;
- Served as a teaching assistant for Java and C++ programming;
- Served as a volunteer PHP developer at the junior enterprise.

- **Visiting Student in Electronic Engineering @ TCD, Ireland**

Sep 2013–May 2014

- Full-term visiting student, earning 60 ECTS with honors;
- Awarded the Brazilian Federal Government's "Science without Borders" (SwB) scholarship;
- Received a Special Prize for the Best Product Branding Video at the Trinity Engineering Student Entrepreneurs of the Year Competition in 2014.

- **Professional Training in Computer Hardware and Network Maintenance @ Microlins, Brazil**

Mar 2007–Sep 2007

- Six-month professional training program in computer hardware, operating systems, and computer networks.